How do RNNs encode mixtures? One network yields distinct geometries

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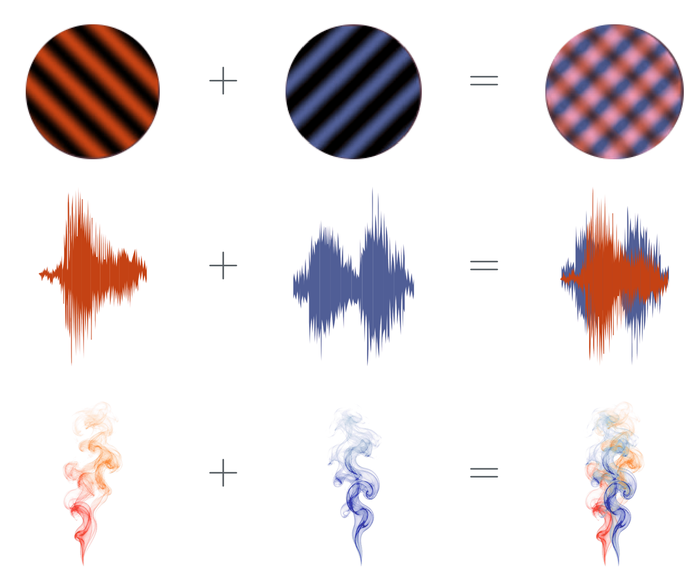
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1 STARTING POINT
Natural inputs often mix simpler components.

Vision
overlapping gratings

Audition
mixed voices

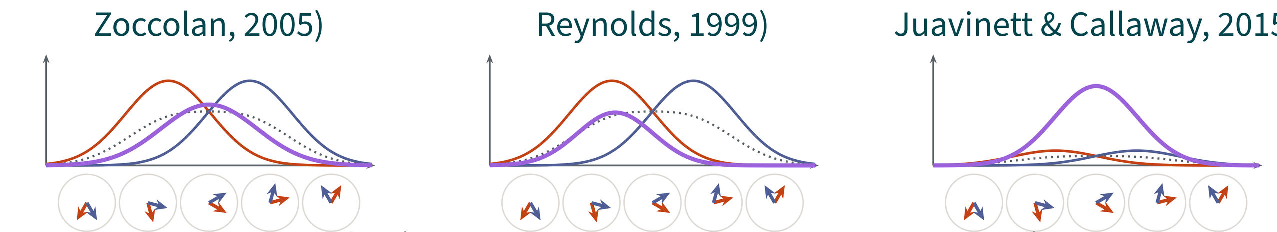
Olfaction
odor blends



2 OBSERVATION
In single neurons, responses are **heterogeneous**.

Average of both component responses	Response closer to one component	Mixture-specific response
(Xiao & Huang, 2015; Zoccolan, 2005)	(Xiao & Huang, 2015; Reynolds, 1999)	(Sadagopan & Wang, 2009; Juavinett & Callaway, 2015)

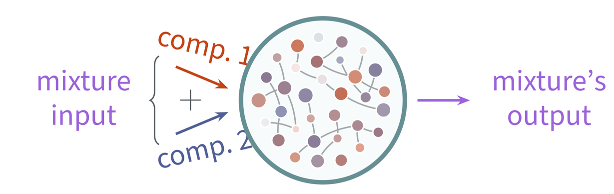
single-neuron response



stimulus (e.g. 2 orientations, alone or superimposed)

— component 1 alone — component 2 alone — mixture ····· linear sum

3 GOAL
Understanding mixture coding in whole networks


mixture input
comp. 1
comp. 2
mixture's output

TWO QUESTIONS

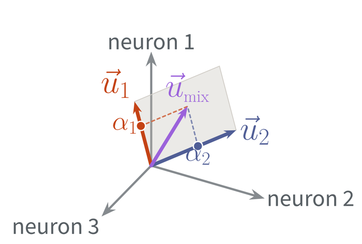
4 QUESTION • Describing codes
Which **distinct codes** for mixtures are possible, *a priori*?

5 APPROACH • Describing codes
Geometry characterizes the code.

Across neurons, a mixture's output is one activity **vector**. Its geometry — its position *relative to* the component outputs — is what varies systematically.

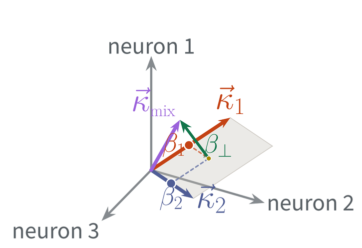
Input vector

$$\vec{u}_{\text{mix}} = \underbrace{\alpha_1\vec{u}_1 + \alpha_2\vec{u}_2}_{\text{linear sum of components}}$$


neuron 1
neuron 2
neuron 3

Output vector

$$\vec{\kappa}_{\text{mix}} = \underbrace{\beta_1\vec{\kappa}_1 + \beta_2\vec{\kappa}_2}_{\text{in the comp. plane}} + \underbrace{\beta_\perp\vec{\eta}_\perp}_{\text{orthogonal to plane}}$$


neuron 1
neuron 2
neuron 3

LINEAR SUM
reference for characterizing geometry

By construction, the mixture input is the linear sum of its components.

The output may deviate from it by:
- a change of balance ($\beta_1{:}\beta_2$ against $\alpha_1{:}\alpha_2$)
- a part out of the plane ($\beta_\perp$)

6 ANSWER • Describing codes
Mixture geometry varies along two independent properties:

BALANCE & EMERGENCE

BALANCE RELATIVE TO THE INPUT

Does the output change the components' balance?

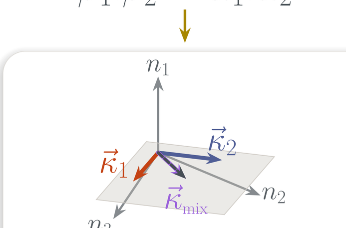
UNBIASED $\beta_1{:}\beta_2 = \alpha_1{:}\alpha_2$

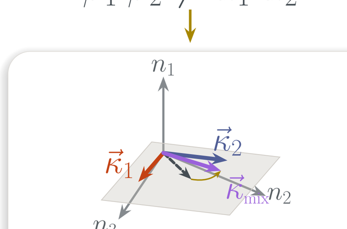
BIASED $\beta_1{:}\beta_2 \neq \alpha_1{:}\alpha_2$

EMERGENCE

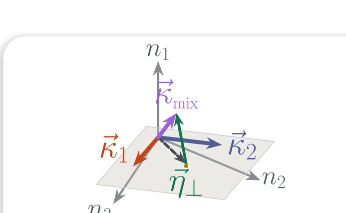
Does the output leave the components plane?

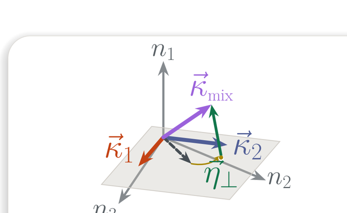
IN PLANE — $\beta_\perp = 0$ →


Unbiased In plane


Biased In plane

OUT OF PLANE — $\beta_\perp \neq 0$ →


Unbiased Out of plane

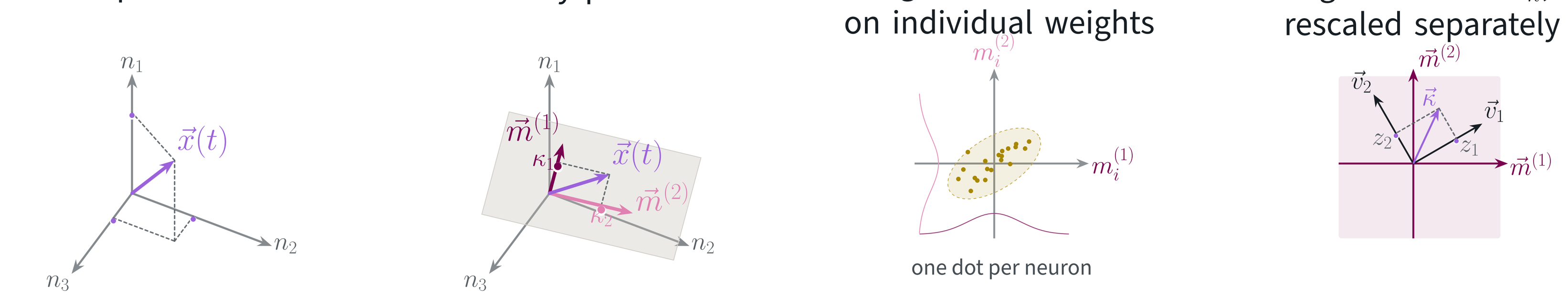

Biased Out of plane

7 QUESTION • Explaining codes
Which network **mechanisms** select the actual code?

8 APPROACH • Explaining codes
A reduced **model** links network structure to geometry. 1

A few biologically plausible assumptions (Mastrogiuseppe & Ostojic, 2018; Beiran et al., 2021; Dubreuil et al., 2022) reduce an established neuron-level model to a few coordinates that capture output geometry.

Standard rate model	Structured connectivity	Gaussian weight distribution	Change of basis
N neurons with external input and recurrent connectivity	sum of R rank-one matrices	in large networks ($N \to \infty$)	eigenvectors of the recurrent covariance
1	2	3	4
N coordinates, one per neuron	R coordinates, one per connectivity pattern $\vec{m}^{(r)}$	outputs depend on weight statistics, not on individual weights	R coordinates, one per eigendirection $\vec{v}_k$, rescaled separately


one dot per neuron

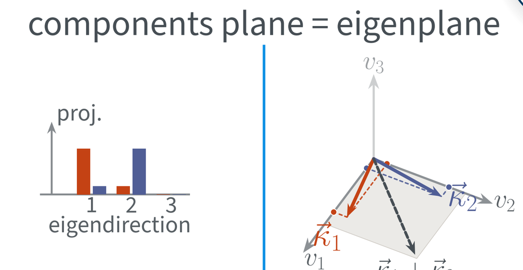
9 ANSWER • Explaining codes
Two successive transformations of the input control the geometry:

FEEDFORWARD PROJECTION & RECURRENT AMPLIFICATION 2 3

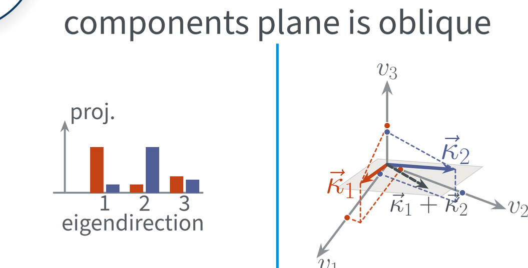
FEEDFORWARD PROJECTION (linear)

How many eigendir. do the components activate jointly?

2 active eigendir. — components plane = eigenplane



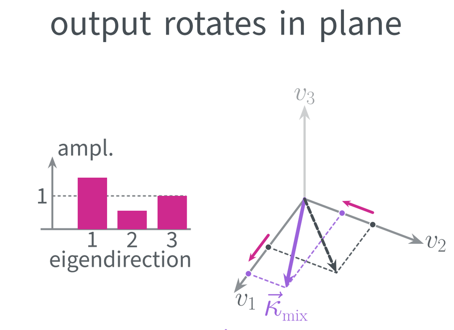
≥ 3 active eigendir. — components plane is oblique



RECURRENT AMPLIFICATION (non-linear)

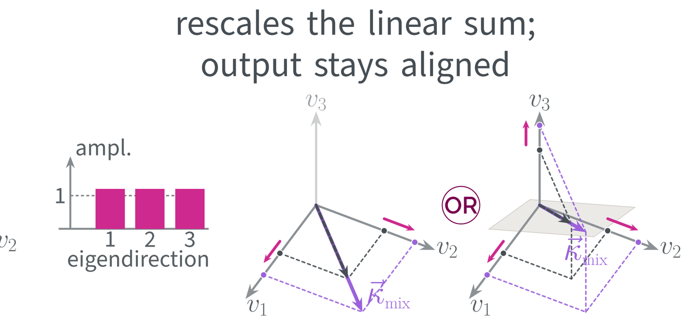
How is the mixture's projection rescaled across the active eigendirections?

Uneven profile with 2 eigendir. — output rotates in plane



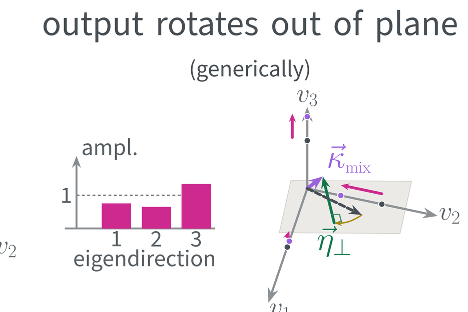
Biased In plane

Flat profile in either case — rescales the linear sum; output stays aligned



Unbiased In plane

Uneven profile with ≥ 3 eigendir. — output rotates out of plane (generically)



Biased Out of plane

10 INTERPRETATIONS
Each geometric property shapes what a readout can infer.

- A **biased** combination can implement attentional selection by **gain control**.
- An **emergent** pattern is **linearly separable** from its components — readable as a distinct identity.

11 PREDICTIONS
The model predicts testable signatures. 4 5 6

- **Biased · Out of plane is the generic outcome**; the other geometries require special inputs or connectivity.
- **Emergence requires dissimilar components**, and grows as they separate.
- **Bias decreases** with stimulus intensity.

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